Report: MA6217 Problem-Based Assignment 4

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2/11/2026

Check all the codes, Simulink files, images, raw data and report at ntulearn or:
github.com/MrOxOrOxen/ntu/tree/ma6217/PBA4.

1 Task 1

The bicycle dynamics are $M\ddot{q} + v_0 C_0 \dot{q} + (K_0 + v_0^2 K_2)q = Hu$, where $q = [\phi, \delta]^T$, $H = [0, 1]^T$ and $u = T$. The lean angle, steering angle, and their angular velocities are selected as the state variables $x = [\phi, \delta, \dot{\phi}, \dot{\delta}]^T$.

Rearranging the dynamic equation gives $\ddot{q} = -M^{-1}(K_0 + v_0^2 K_2)q - M^{-1} v_0 C_0 \dot{q} + M^{-1} H u$. Using the relationship between angles and angular velocities, the system is converted into the first-order state-space form $\dot{x} = Ax + Bu$ and $y = Cx + Du$, where

$$A = \begin{bmatrix} 0_{2\times 2} & I_2 \\ -M^{-1}(K_0 + v_0^2 K_2) & -M^{-1} v_0 C_0 \end{bmatrix}, B = \begin{bmatrix} 0_{2\times 1} \\ M^{-1} H \end{bmatrix}.$$

Since the system output is the steering angle δ, $C = [0, 1, 0, 0]$ and $D = 0$.

Using MATLAB, the following matrices are obtained at $v_0 = 5\,\mathrm{m/s}$:

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 8.7611 & 23.2100 & -0.6370 & 2.1486 \\ -14.9477 & 29.1529 & -17.2465 & -12.9089 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 0 \\ 0.2922 \\ 7.9109 \end{bmatrix}.$$

The open-loop eigenvalues are calculated using the MATLAB command eig(A):

$$\lambda(A) = \{1.2964 \pm 4.6029j, -1.8426, -14.2961\}$$

Since one pair of complex conjugate eigenvalues has a positive real part 1.2964, the system contains unstable modes. Therefore, the bicycle is open-loop unstable at a forward speed of $5\,\mathrm{m/s}$, and we need a feedback control to stabilize the system.

2 Task 2

Firstly, the controllability matrix is evaluated using MATLAB: $\mathrm{rank}([B, AB, A^2B, A^3B]) = 4$. Since the rank equals the number of states, the system is fully controllable, and pole placement can be achieved using state feedback.

The state feedback control law is defined as $u = -Kx$, giving the closed-loop system $\dot{x} = (A - BK)x$.

Using the MATLAB function place(A,B,p), the closed-loop poles are assigned to -2, -10, and $-1 \pm j$. The feedback gain matrix is obtained as:

$$K = \begin{bmatrix} 0.3247 & 8.4043 & -1.3455 & 0.1071 \end{bmatrix}.$$

The closed-loop eigenvalues are calculated as:

$$\lambda(A - BK) = \{-10, -2, -1 \pm j\}.$$

The calculated eigenvalues match the desired pole locations. Since all closed-loop eigenvalues have negative real parts, the closed-loop system is asymptotically stable. Therefore, the designed state feedback controller successfully stabilizes the bicycle and satisfies the specified pole placement requirements.

3 Task 3

The state feedback controller designed in Task 2 is applied using the control law $T = -Kx + k_r r$, where $r = 0.002\,\text{rad}$ is the steering reference step input and k_r is the reference gain.

The system response is simulated in MATLAB over 15 seconds, as shown in Figure 1, whose steering angle δ approaches the reference value with a small overshoot at approximately 4 s and settles near 0.002 rad after about 6 s, with negligible steady-state error.

Meanwhile, the lean angle ϕ converges to approximately $-0.0048\,\text{rad}$, and the steering torque T settles at approximately $-0.0165\,\text{N}\cdot\text{m}$, without sustained oscillations.

Therefore, the designed full-state feedback controller stabilizes the system and successfully tracks the steering reference.

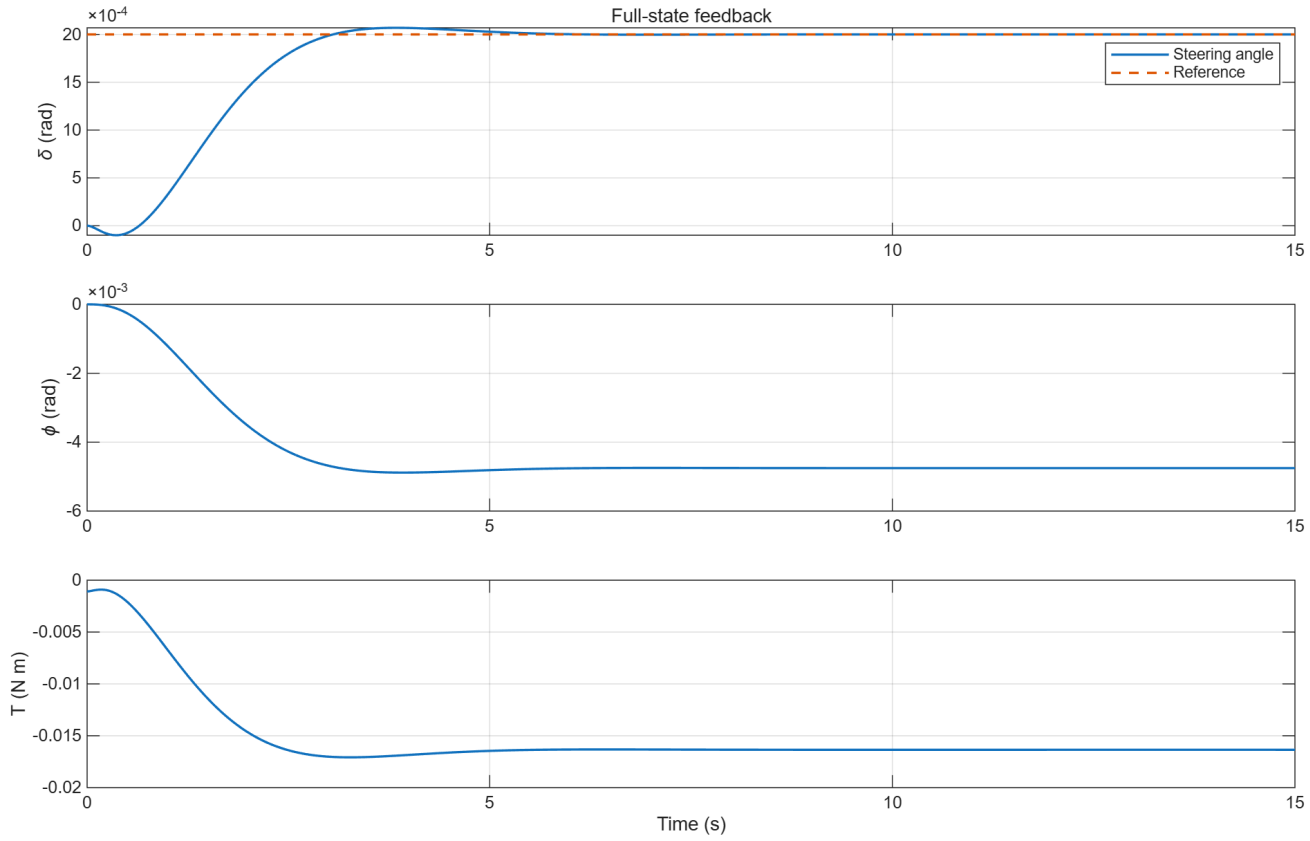


Figure 1: Step response of the bicycle with full-state feedback control.

4 Task 4

For a linear system, the state-space equations are $\dot{x} = Ax + BT$ and $y = Cx$.

In Task 3, the full-state feedback control law is $T = -Kx + k_r r$. However, when not all states can be directly measured, a state observer is required to estimate the system states for feedback control.

First, the state prediction is obtained from the system model $\dot{\hat{x}} = A\hat{x} + BT$. To correct the prediction error, the difference between the measured output y and the estimated output $\hat{y} = C\hat{x}$ is introduced. Therefore, the observer equation becomes:

$$\dot{\hat{x}} = A\hat{x} + BT + L(y - C\hat{x}).$$

where $\hat{x}$ is the estimated state and L is the observer gain matrix.

Defining the estimation error as $e = x - \hat{x}$, subtracting the observer equation from the system equation gives

$$\dot{e} = \dot{x} - \dot{\hat{x}} = A(x - \hat{x}) - LC(x - \hat{x})$$

Thus, the error dynamics are: $\dot{e} = (A - LC)e$. Therefore, the estimation error converges to zero if all eigenvalues of $A - LC$ have negative real parts.

In this task, only the steering angle is measured, so $C = [0, 1, 0, 0]$. The observability of the system is first checked using MATLAB:

$$\operatorname{rank}(O) = \operatorname{rank}\begin{bmatrix} C \\ CA \\ CA^2 \\ CA^3 \end{bmatrix} = 4.$$

Since the rank equals the number of states 4, the system is fully observable.

Using the MATLAB function place, the observer poles are assigned to -8, -16, and $-4 \pm j$. The observer gain matrix is obtained as:

$$L = \begin{bmatrix} -23.1167 \\ 18.4541 \\ -3.5407 \\ 79.6573 \end{bmatrix}.$$

The eigenvalues of $A - LC$ are calculated as $\lambda(A - LC) = \{-16, -8, -4 \pm j\}$, which match the desired pole locations.

The feedback gain K from Task 2 is retained, and the controller uses the estimated states $T = -K\hat{x} + k_r r$.

Combining the actual system and the observer gives:

$$\begin{bmatrix} \dot{x} \\ \dot{\hat{x}} \end{bmatrix} = \begin{bmatrix} A & -BK \\ LC & A - BK - LC \end{bmatrix} \begin{bmatrix} x \\ \hat{x} \end{bmatrix} + \begin{bmatrix} Bk_r \\ Bk_r \end{bmatrix} r.$$

The poles of the combined system consist of the feedback controller poles and the observer poles. Since all poles have negative real parts, the combined system is asymptotically stable.

The system is simulated in MATLAB with a steering reference step input of $r = 0.002\,\text{rad}$.

Figure 2 shows the actual and estimated values of all four states. Initial estimation differences are observed, but the estimated states gradually approach the actual states. Eventually, the corresponding curves nearly overlap.

Figure 3 shows the estimation errors of all four states. All errors become negligible within approximately 2 s, verifying the effectiveness of the observer.

Figure 4 compares the responses of full-state feedback and observer-based feedback. Due to the initial estimation error, observer-based feedback produces larger transient deviations. The steering angle reaches approximately $-0.014\,\text{rad}$, and the peak steering torque is approximately $0.115\,\text{N}\cdot\text{m}$. However, as the estimation error converges, the system gradually stabilizes and eventually tracks the 0.002 rad steering reference.

Therefore, the designed state observer provides effective state estimation and achieves stable output feedback control. However, the initial estimation error has a significant influence on the transient response.

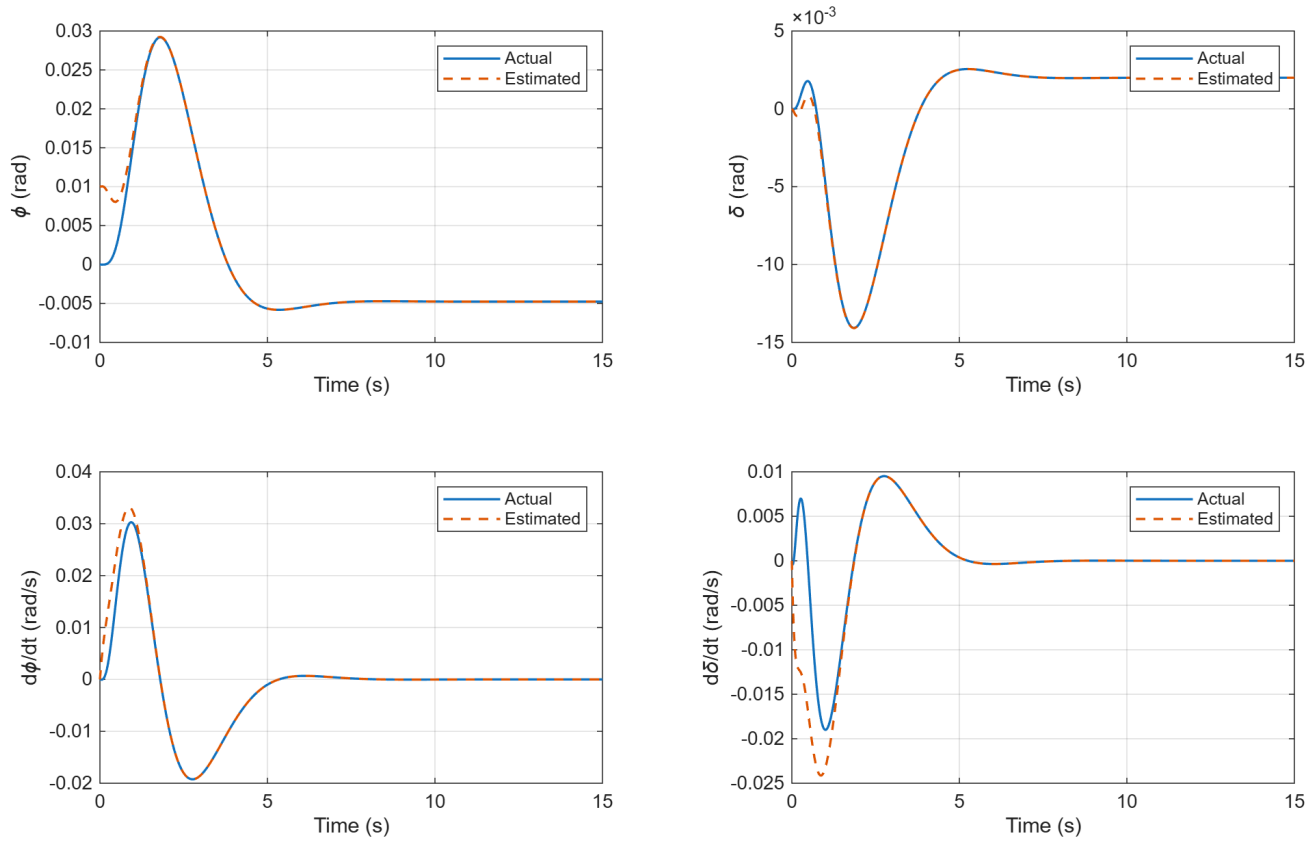


Figure 2: Actual and Estimated State Responses

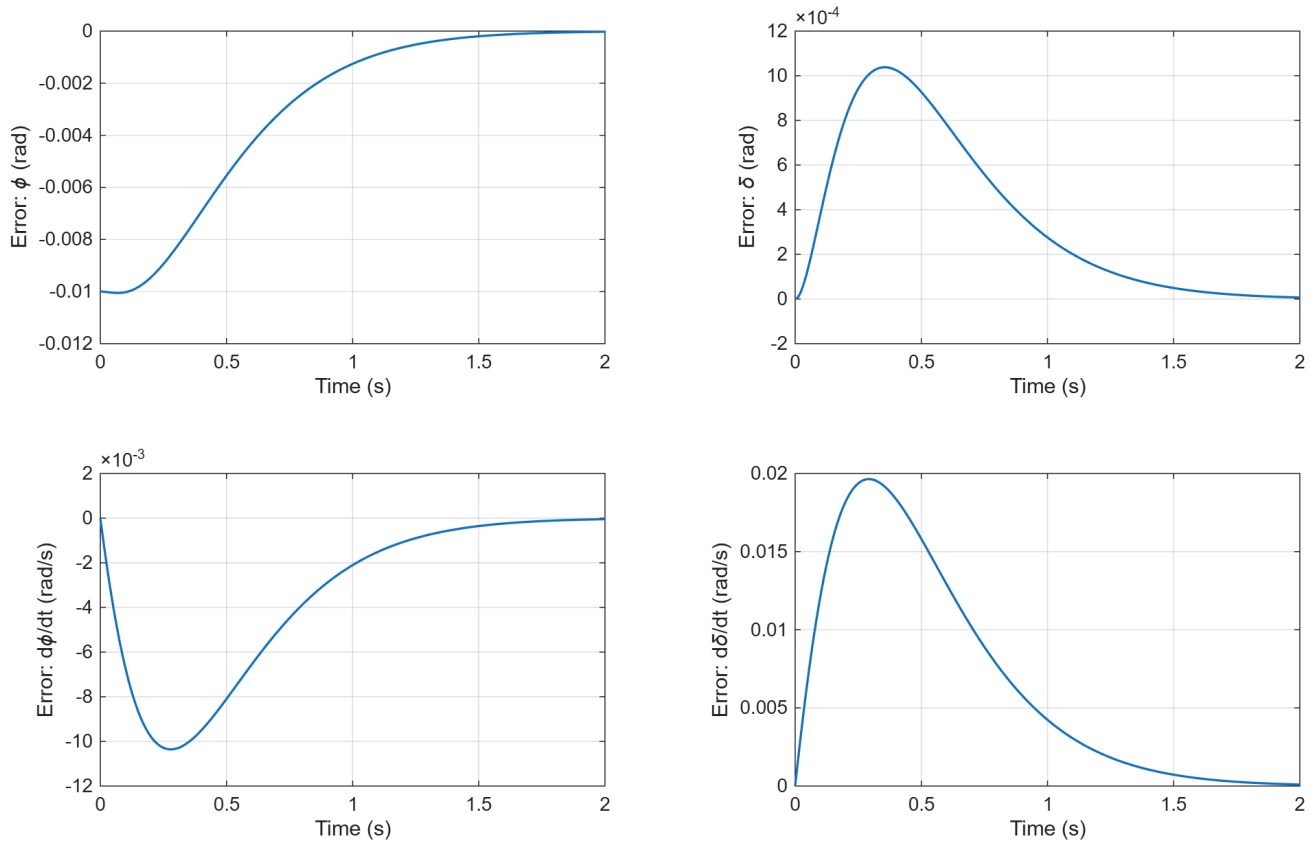


Figure 3: State Estimation Errors

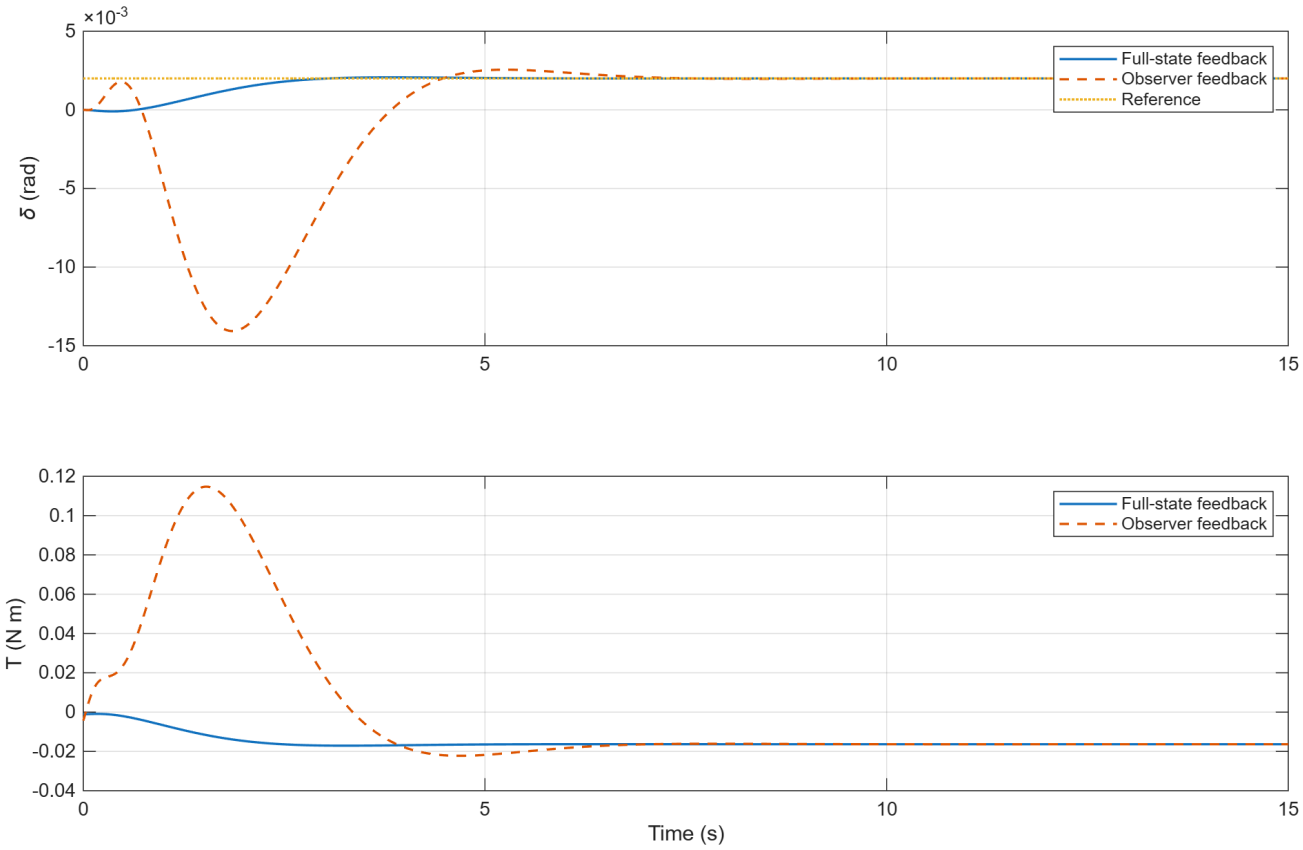


Figure 4: Full-State and Observer-Based Feedback Comparison

5 Task 5

From Task 4, the estimation error dynamics are given by $\dot{e} = (A - LC)e$.

Using the observer-based feedback control law $T = -K\hat{x} + k_r r$ and $\hat{x} = x - e$, the combined system becomes:

$$\begin{bmatrix} \dot{x} \\ \dot{e} \end{bmatrix} = \begin{bmatrix} A - BK & BK \\ 0 & A - LC \end{bmatrix} \begin{bmatrix} x \\ e \end{bmatrix} + \begin{bmatrix} Bk_r \\ 0 \end{bmatrix} r.$$

Therefore, the poles of the combined system consist of the eigenvalues of $A - BK$ and $A - LC$. According to the separation principle, changing the observer poles does not affect the state feedback poles, but it changes the estimation error convergence and influences the transient response through BKe.

To investigate the effect of observer pole placement, the observer poles from Task 4 are scaled by a factor α:

$$p_o(\alpha) = \alpha\{-8, -16, -4 \pm j\}.$$

We tested five values of $\alpha = 0.25, 0.5, 1, 2, 4$ in MATLAB. The observer gain L is recalculated using the place function for each configuration. The state feedback gain K, steering reference $r = 0.002\,\text{rad}$, and initial conditions are kept unchanged.

Figure 5 compares the steering responses, normalized estimation errors, and steering torques for different observer poles.

Figure 6 further compares the steering responses and control torques for $\alpha = 1, 2, 4$.

The performance metrics calculated in MATLAB are summarized in Table 1. The estimation error settling time t_e is defined using state tolerances of $[10^{-4}, 10^{-4}, 10^{-3}, 10^{-3}]$, while the steering angle settling time t_s is calculated using a 2% tolerance of the reference value. Both settling times require the corresponding errors to remain within their tolerances.

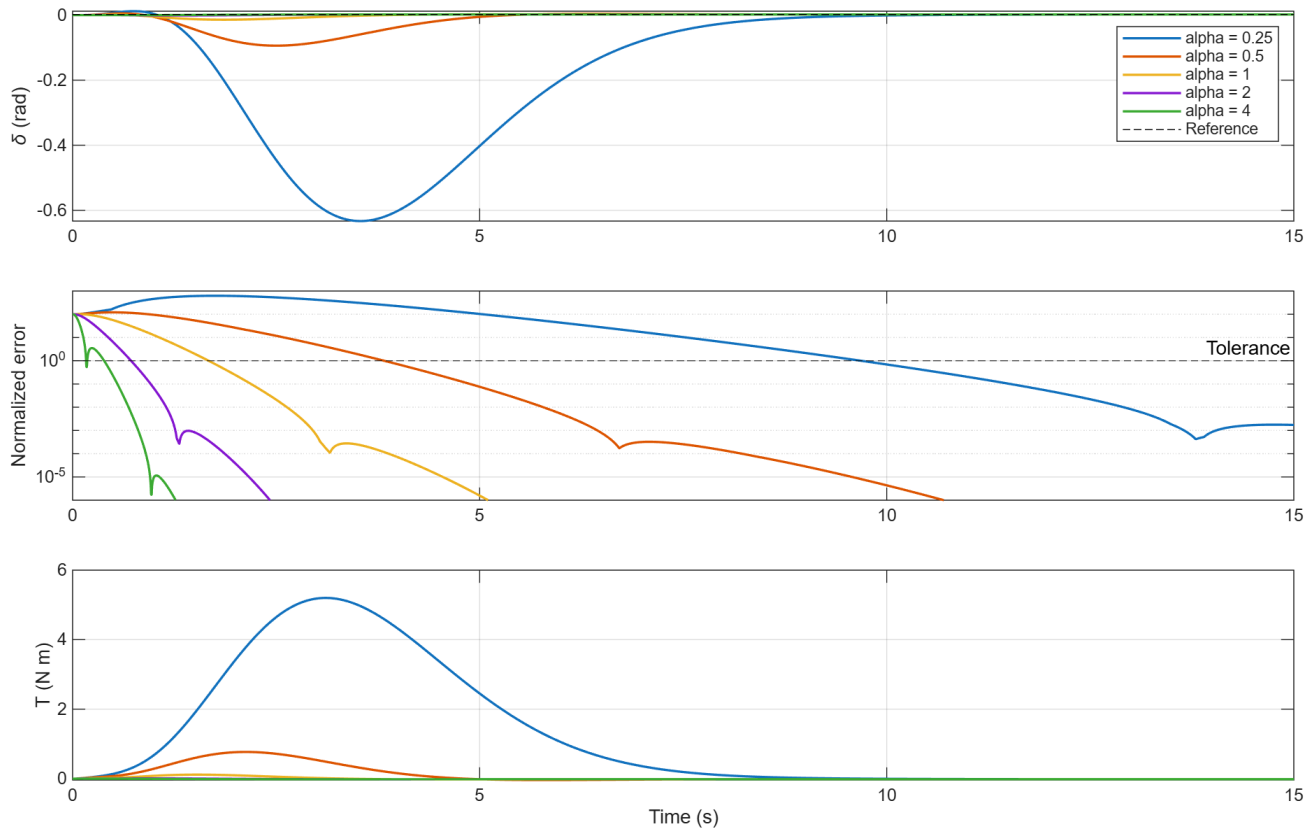


Figure 5: Effects of observer pole scaling on steering response, estimation error, and control torque.

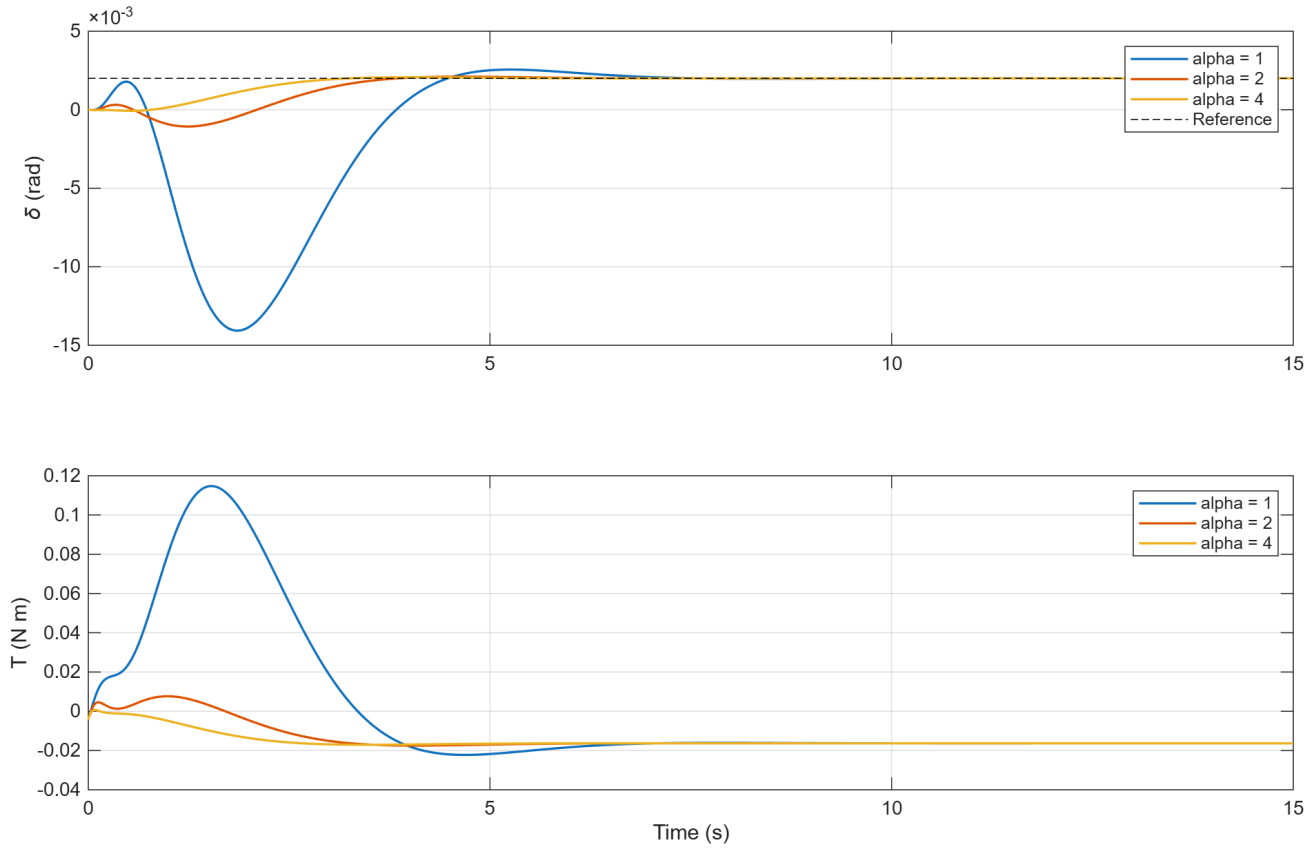


Figure 6: Steering responses and control torques for faster observer configurations.

Table 1: Performance comparison for different observer pole configurations.

α	t_e (s)	t_s (s)	Peak Error (rad)	Peak $\lvert T\rvert$ (N m)	$\lVert L\rVert_2$
0.25	9.683	13.501	0.63511	5.2007	89.26
0.5	3.826	10.636	0.095894	0.7685	46.516
1	1.671	7.231	0.016068	0.11473	85.046
2	0.719	5.852	0.003069	0.017473	765.8
4	0.387	4.898	0.002068	0.017064	5543.6

The results show that increasing α moves the observer poles further into the left-half plane and significantly reduces the estimation error settling time. Faster estimation also reduces the influence of estimation errors on the control torque, improving the steering response.

For $\alpha = 0.25$, slow estimation produces large steering deviations and control torque. When α increases to 2, the peak tracking error decreases to approximately 0.0031 rad, while the peak torque decreases to approximately 0.0175 N m.

However, increasing α further to 4 provides only limited improvement in control performance, while the observer gain norm increases from 765.8 to 5543.6. Such a large observer gain may increase sensitivity to measurement noise and model uncertainties, although these effects are not directly tested in this simulation.

Therefore, observer poles should be selected to provide sufficiently fast estimation error convergence without excessively large observer gains. The selection should balance estimation accuracy, steering tracking performance, and control effort.

And for this simulation, we finally choose $\alpha = 2$, which can provide a reasonable compromise between control performance and observer gain magnitude.